

Quesion

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a) A full-wave rectifier is to supply a load of 650mA at 5V dc with a ripple that must be less than 700mV. Specify the elements of the rectifier using a fuller capacitor and a 78xx voltage regulator integrated chip (Take  $f = 50\text{Hz}$  and mains voltage 220V)

b) Draw the full circuit diagram

Solution

EEE=434

a) Choose 7805 IC. 5V, 1A.  
voltage input from 8V to 35 volts

Choose 28 volts as input (dc voltage)

b) Choosing the working voltage and capacitance of a capacitor  
from  $V_r = \frac{I_{dc}}{4fc} \rightarrow \text{Ripple voltage}$

$$I_{dc} = 650\text{mA}$$

$$V_r = 0.7\text{V}$$

Substituting the values yields

$$0.7 = \frac{0.65}{4 \times 50 \times c}$$

Working voltage  
15 16V

$$0.7 \times 4 \times 50 \times c = 0.65$$

$$1400c = 0.65$$

$$c = \frac{0.65}{1400} = 0.0004642857$$

$$\frac{46400\mu\text{F}}{100} = 5000\mu\text{F}$$

$$92857\mu\text{F}$$

The standard value and close to the answer is  
and the working voltage is

Choosing the transformer

$$V_m = V_{dc} + \frac{I_{dc}}{4f_c}, \quad \frac{0.56}{4 \times 50 \times c}$$

$$V_m = 28.7 + 0.7 = 28.7$$

PIV = peak inverse voltage

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{27.7}{\sqrt{2}} = \frac{27.7}{1.414} = 19V$$

Hence a 18-0-18 would be used (or any slightly higher standard voltage transformer).  
Transformer specs:  $V = 19V$  (or any higher standard value)  
 $I = 650mA$

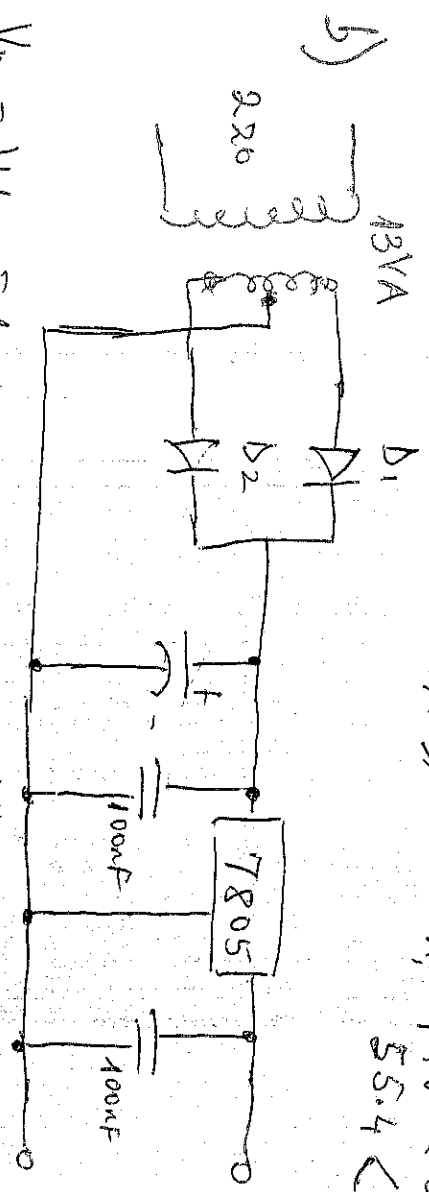
$$VA = \frac{VR}{\eta} = 19 \times 650mA = 12.35$$

choosing diode

d) 1N4001 1N4001

choice. Since  $I = 1A > 650mA$

$\approx 13VA$  or next slightly higher standard value.  
 $PIV = 2V_m = 2 \times 28.7 = 55.4 < PIV \text{ of } 1N4001$



$$V_m = V_{dc} + \frac{I_{dc}}{4f_c} = \frac{0.65}{4 \times 50 \times c}$$

$$V_m = 28.7 + 0.7$$

$$28.7 + 1.7 \text{ (full wave)}$$

$$28.7 + 0.7 \text{ (full wave with)}$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$I = 650mA$$

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{29.4}{\sqrt{2}} = 20.88V \approx 21V$$

$$Power \times V \times I = 21 \times 650 = 13.65VA$$

0.7 Silicon  
0.3 Germanium

Group A

V<sub>cc</sub> = 240  
@ 95

01 - 40

Construction ok.

Testing ok.

07 absent  
ok.

V<sub>n</sub>

R.

12V

400m

Group B

41 - 20

Construction ok.

Testing ok.

$I = \frac{P}{400}$

$\frac{9}{400}$

= 0.0225

22mA

15 - 20mA

Group C

24

20 ok.

30

Construction ok.

Testing ok.

24 ok.

25 ok.

30 ok.

(22 to 100)

R =

9 - 2.7

20mA.

0.315 x 10<sup>3</sup>

Group D

31

14/10

43

34 ok.

39 ✓

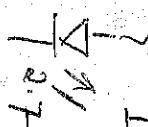
41 ✓

43 ✓

9 - 2.7

9

$\frac{9-2.7}{I}$



9.22

20mA

315mA

